

IRONWOMAN · RANDOMISED TRIAL

What iron therapy *really* *does* for tired, iron- deficient women.

We put it to the test in a double-blind randomised trial. The results are not what most people expect.

Swipe



Including the parts that did not work.

IRONWOMAN · WHAT WE FOUND

What changed. *What did not.*

CHANGED WITH IRON

-  Serum ferritin
-  Serum iron
-  Transferrin saturation
-  Running economy
-  Fatigue scores

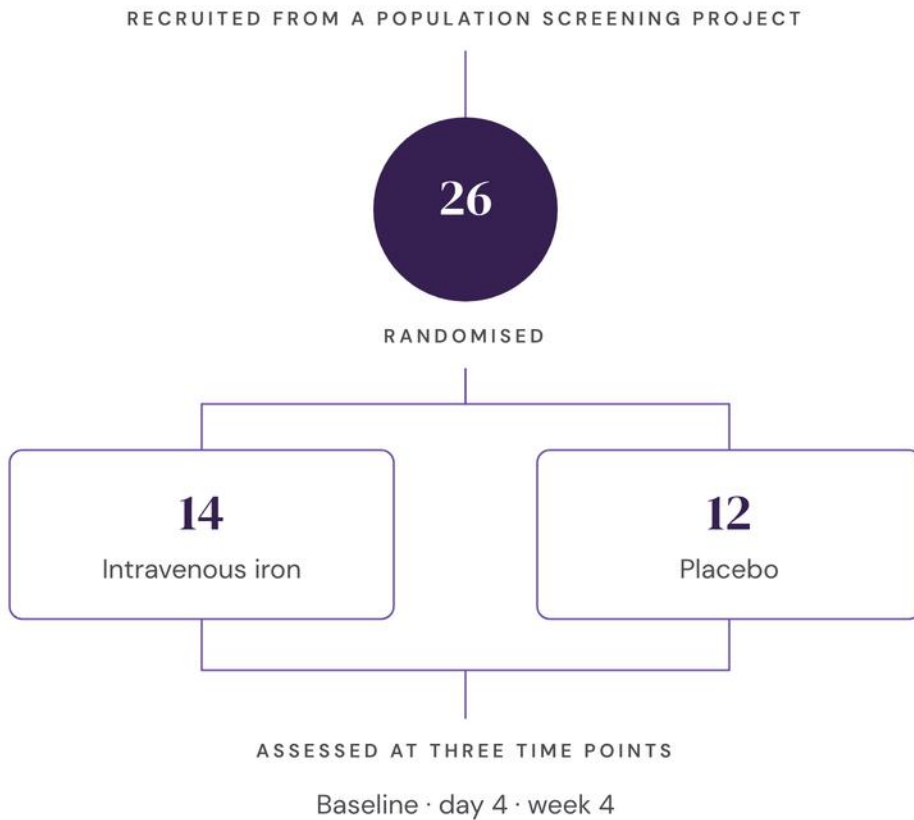
DID NOT CHANGE

-  Peak oxygen uptake
-  Lactate threshold
-  Haemoglobin mass
-  Mood states
-  Quality of life

*It corrected iron status, economy and fatigue.
Peak capacity did not move.*

IRONWOMAN · WHAT WE DID

A double-blind randomised trial.

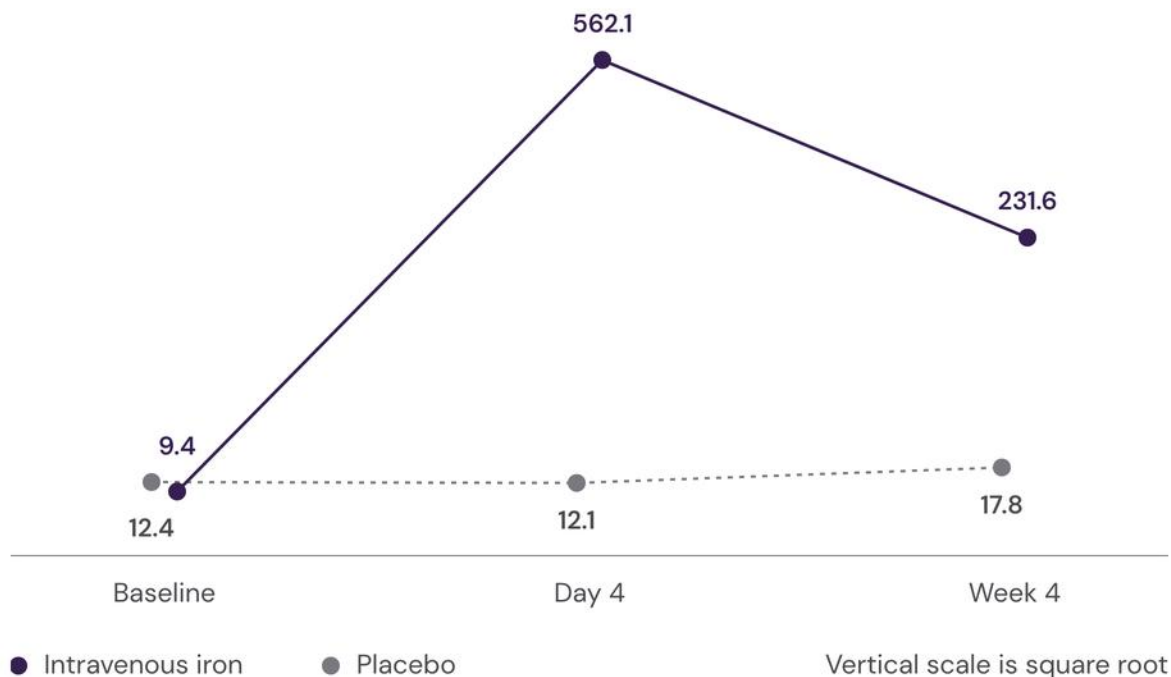


Twenty-six iron-deficient, non-anaemic, recreationally active women.

IRONWOMAN · IRON STATUS

Iron therapy *corrects the deficiency.*

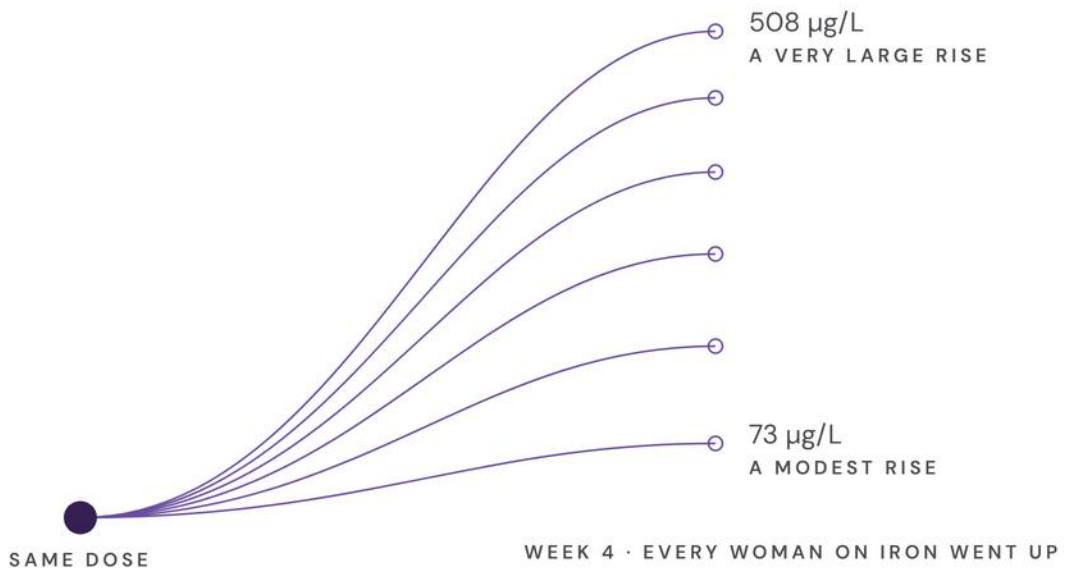
Serum ferritin, $\mu\text{g/L}$



Four weeks on, stores were still many times where they started.

IRONWOMAN · THE SPREAD

One dose. *Twenty-six answers.*



Everyone responded. The size of the response was not the same.

IRONWOMAN · WHAT IT MEANS

What to take from *a null result*.

Treating it is worth doing for how you feel.

Fatigue and running economy improved. Those are the things people actually notice day to day.

It is not a performance drug.

Peak capacity did not move. Correcting a deficiency is not the same as an upgrade.

Your response will not be the average response.

Individual responses varied widely, which is the argument for retesting rather than assuming.

The honest version is more useful than the hopeful one.

IRONWOMAN · FOR A WORKFORCE

What this means if you *screen people*.

If you are screening a workforce for iron deficiency, this trial tells you what your program should be measuring.

MEASURE	Ferritin, and fatigue
EXPECT	Function and how people feel to shift
DO NOT USE	A fitness test as your outcome
DO NOT ASSUME	An average response for any individual

A null result on the wrong outcome is a specification for the right one.

ONE NUMBER

The test that would have found you.

If you are tired and you have been told your bloods are normal, a full blood count is not the test that would show this.

Ask for ferritin.

One number is the difference between "you are fine" and an answer.

READ IT YOURSELF

The whole thing is open to check.

Dugan C, Peeling P, Buissink P, MacLean B, Lim J, Jayasuriya P, Richards T. Effect of intravenous iron therapy on exercise performance, fatigue scores and mood states in iron-deficient recreationally active females of reproductive age: a double-blind, randomised control trial. *British Journal of Sports Medicine* 2025;59(13):921-930.

doi 10.1136/bjsports-2024-108240 · ACTRN12620001357943

Every number in this carousel is in that paper.